

The Agent's Signature: Identity & Law in the Age of AI

Illustrated Edition with Diagrams

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Chapter 1

The Agent's Signature

1.1 Identity, Authorization & Law in the Age of Autonomous AI

A Comprehensive Technical & Legal Framework

By Mauricio A. Fernandez Fernandez

“In the age of autonomous agents, every transaction is a legal act, every algorithm a potential fiduciary, and every cryptographic signature carries the weight of centuries of contract law.”

Chapter 2

Preface: Why This Book Matters Now

The year is 2025. Every major technology company has deployed thousands of AI agents handling customer service, code generation, supply chain optimization, and financial transactions. Yet a fundamental question remains unanswered:

When an AI agent signs a contract, who is liable?

This isn't a hypothetical. Bloomberg estimates that over \$4 trillion in transactions are now initiated by autonomous systems annually. McKinsey projects this will reach \$25 trillion by 2030. Yet our authorization infrastructure—OAuth 2.0, originally designed for “Login with Google” buttons—was never built for agents that act, not just access.

This book provides the definitive technical and legal framework for building **trustworthy autonomous agents**. It is based on the **AgentAuth** open-source implementation (v1.0.0, December 2025), which implements the AAP-001 and AAP-002 protocols.

Chapter 3

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Chapter 4

Part I: Foundations

Chapter 5

Chapter 1: The OAuth Trap — Why Access Tokens Aren't Enough

5.1 1.1 The Fundamental Distinction: Access vs. Authority

Every digital system today operates on a simple premise: **authenticate** the user, then **authorize** their actions. OAuth 2.0 perfected this model for web applications. But it has a fatal flaw.

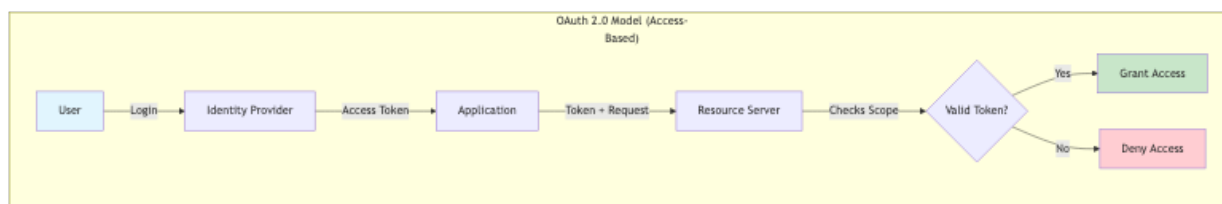


Figure 5.1: Diagram 1

Figure 1: Architecture diagram

Figure 1.1: The OAuth 2.0 access model — simple, effective, but legally blind

This model answers: *“Can this entity access this resource?”*

It does **not** answer: - *“Is this entity legally empowered to bind its principal?”* - *“Does this action fall within fiduciary duties?”* - *“Who is liable if this transaction goes wrong?”*

Consider the legal distinction:

Concept	Technical Analog	Legal Implication
Access Token	House key	Can unlock door; no authority to sell house
Delegation Token	Power of Attorney	Can act on behalf of principal; legally binding
PoA Token	Notarized Limited Power of Attorney	Cryptographically verifiable; scope-limited; auditable

5.2 1.2 Real-World Failure Modes

5.2.1 Case Study 1: The Autonomous Procurement Agent

A manufacturing company deploys an AI agent to handle routine procurement. The agent has an OAuth token with `procurement:write` scope.

Incident: The agent identifies a 40% price reduction on specialty chemicals from a supplier in Belarus (sanctioned jurisdiction). It executes the \$2.3M purchase order.

Technical Authorization: Valid token, valid scope **Legal Authorization:** Violated OFAC sanctions **Outcome:** \$18M fine, criminal investigation

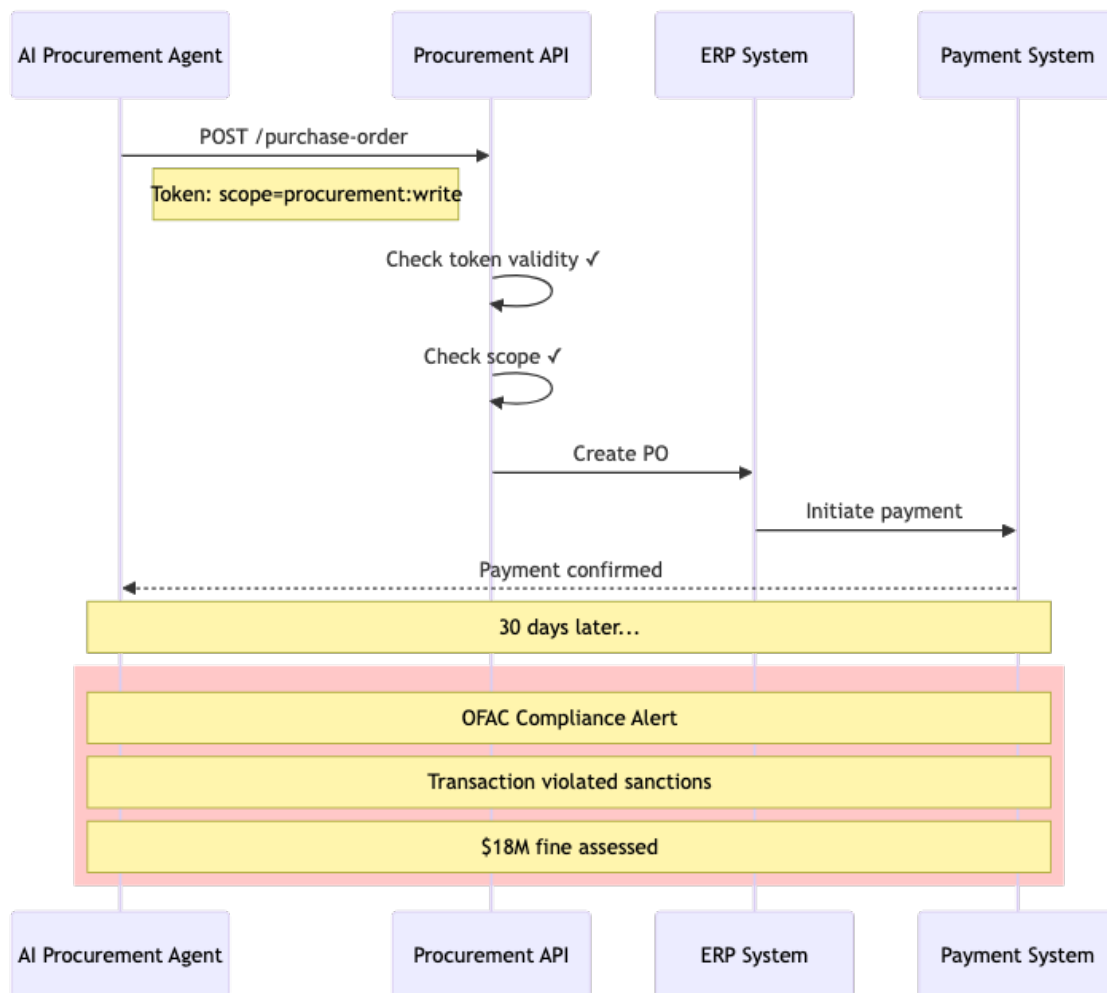


Figure 5.2: Diagram 2

Figure 2: Architecture diagram

Figure 1.2: OAuth authorization succeeds; legal authorization fails catastrophically

5.2.2 Case Study 2: The Healthcare AI Advocate

An elderly patient designates an AI agent as their healthcare advocate. The agent has OAuth access to their medical records.

Incident: Patient becomes incapacitated. Agent approves experimental surgery.

Technical Authorization: Valid API access **Legal Authorization:** No valid healthcare proxy on file **Outcome:** Surgery performed without legal consent; hospital faces liability

5.3 1.3 The Solution: Proof of Authorization (PoA)

AgentAuth introduces a new primitive: the **Proof of Authorization Token (PoA)**. Unlike an access token, a PoA carries:

1. **Principal Identity:** Cryptographically verified human/organizational identity
2. **Agent Identity:** Cryptographically bound software identity
3. **Scope Constraints:** What the agent can and cannot do
4. **Liability Limits:** Maximum financial exposure
5. **Jurisdictional Binding:** Applicable law
6. **Revocation Path:** How authority can be terminated

Figure 3: Architecture diagram

Figure 1.3: The AgentAuth PoA model — legal authority, not just access

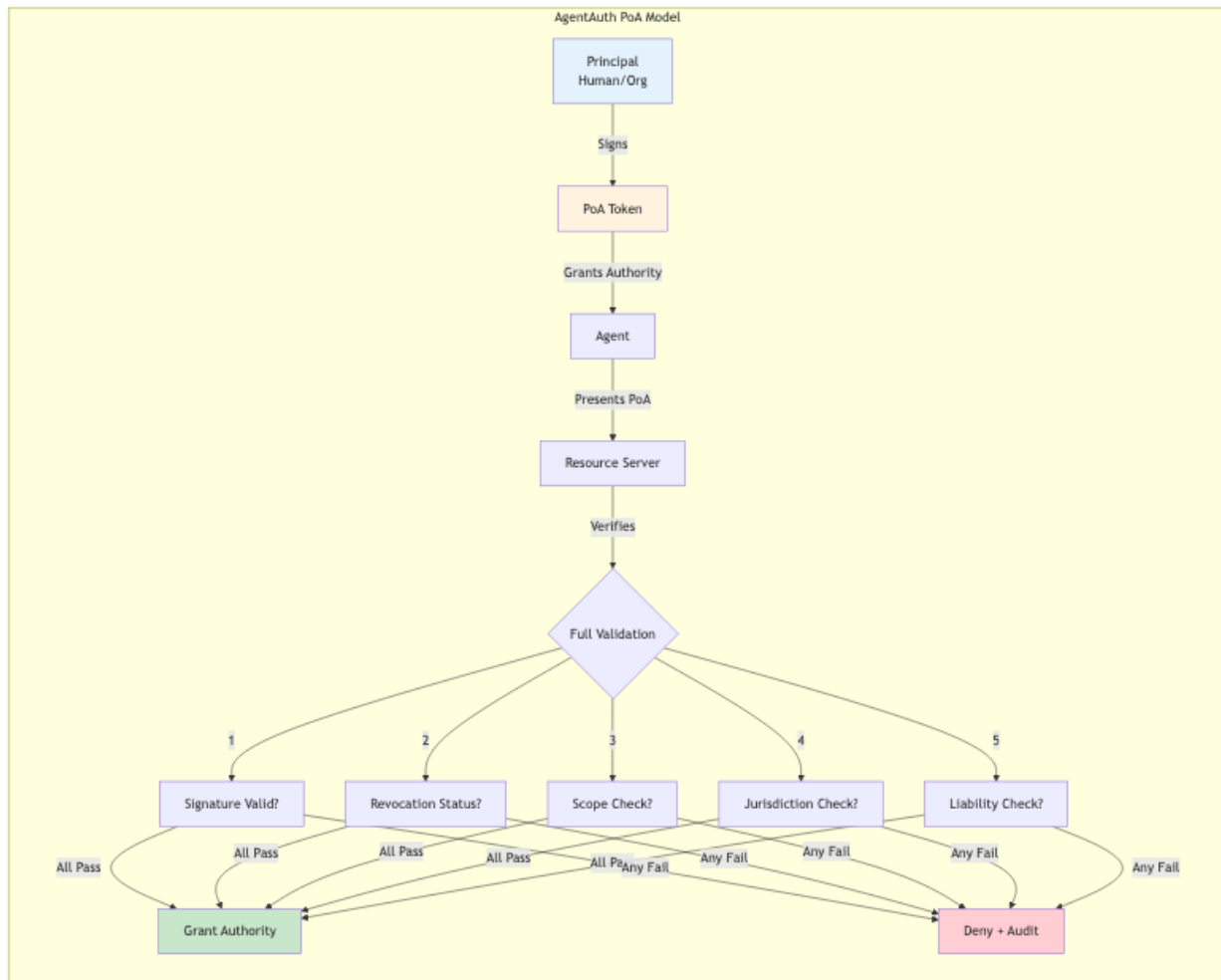


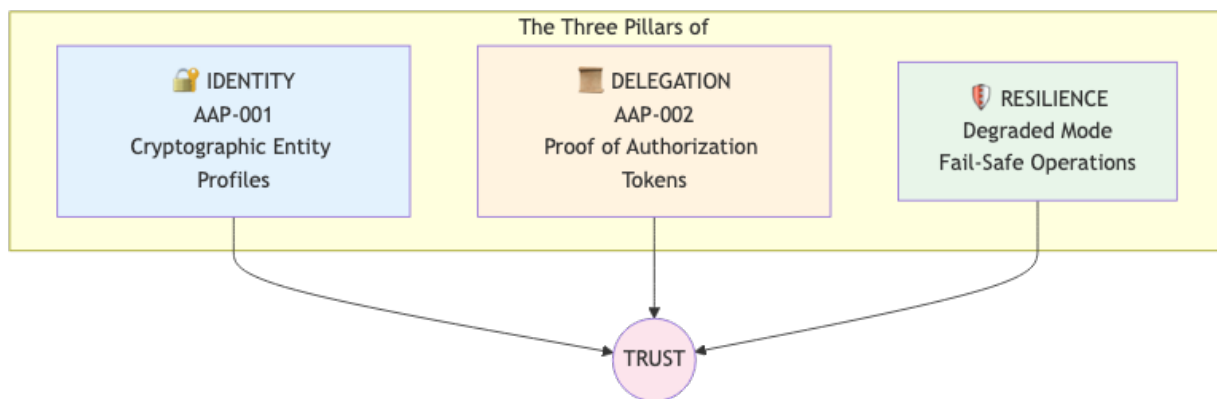
Figure 5.3: Diagram 3

Chapter 6

Chapter 2: The Architecture of Trust

6.1 2.1 The Three Pillars

AgentAuth is built on three foundational pillars:



```

"entity_type": "autonomous_agent",
"public_key": {
  "algorithm": "Ed25519",
  "key": "MCowBQYDK2VwAyEAp6s7p8K2H3R4T5u6V7w8X9..."
},
"legal_metadata": {
  "owning_organization": {
    "name": "Acme Corporation",
    "jurisdiction": "DE",
    "registration": "HRB 123456",
    "register": "Amtsgericht München"
  },
  "agent_classification": "L3_AUTONOMOUS",
  "liability_cap_eur": 1000000
},
"authorization_chain": {
  "root_authorizer": "urn:agentauth:entity:acme-board-resolution-2024",
  "chain_depth": 2,
  "chain_signature": "..."
},
"validity": {
  "not_before": "2025-01-01T00:00:00Z",
  "not_after": "2026-01-01T00:00:00Z"
},
"signature": "..."
}

```

Example 2.1: An AgentAuth Entity Profile for an autonomous procurement agent

6.2.2 The Authorization Chain

Authority flows from humans to agents through a verifiable chain:

Figure 5: Architecture diagram

Figure 2.2: Authority delegation chain — from corporate governance to autonomous agent

Each link in the chain: - Is cryptographically signed by the delegator - Contains the scope of delegated authority - Cannot exceed the delegator's own authority - Can be independently verified

6.3 2.3 Pillar 2: Delegation (AAP-002)

6.3.1 The Proof of Authorization Token

The PoA token is the primary artifact for agent authorization:

Figure 6: Architecture diagram

Figure 2.3: PoA Token structure — comprehensive authorization metadata

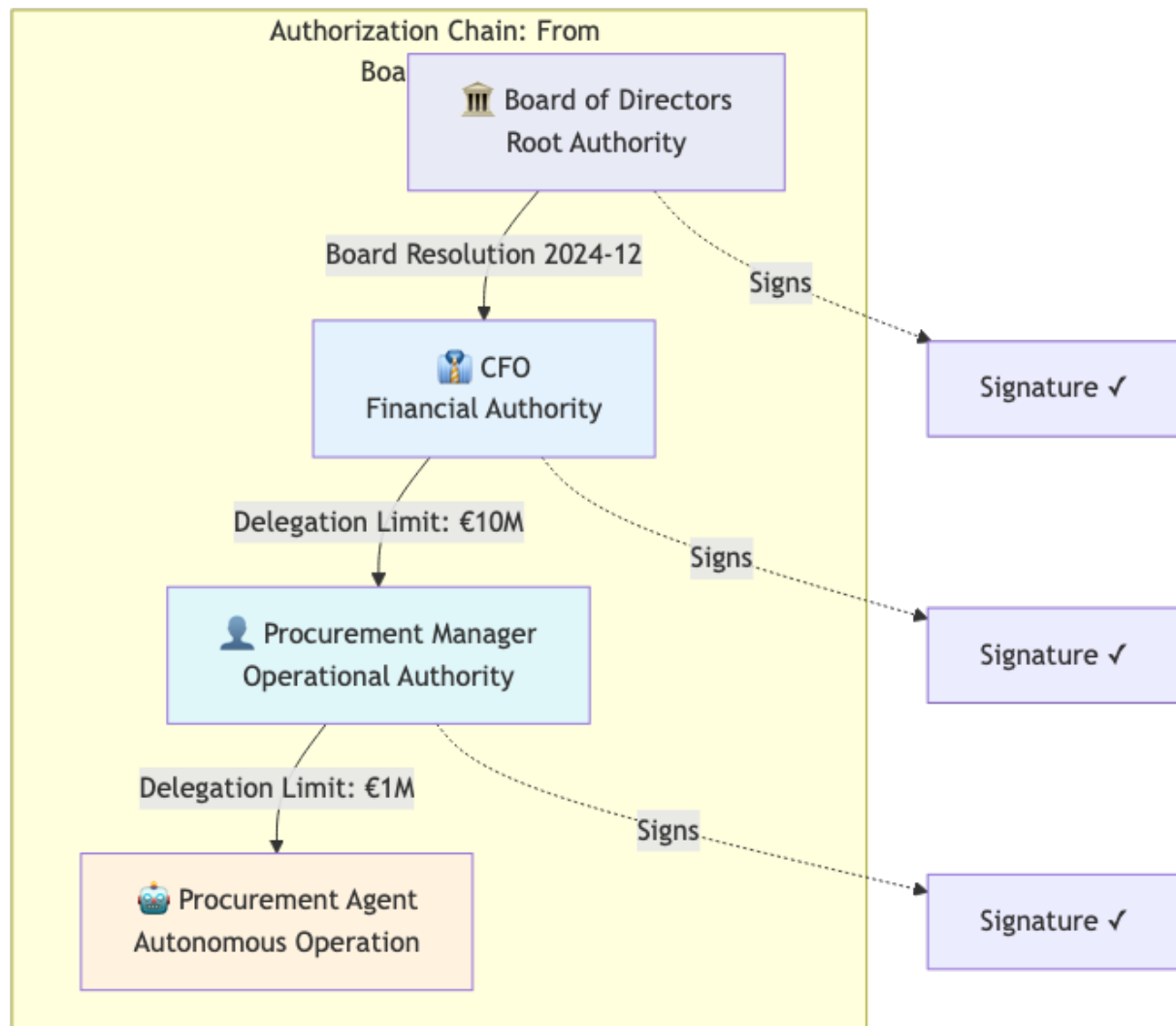


Figure 6.2: Diagram 5

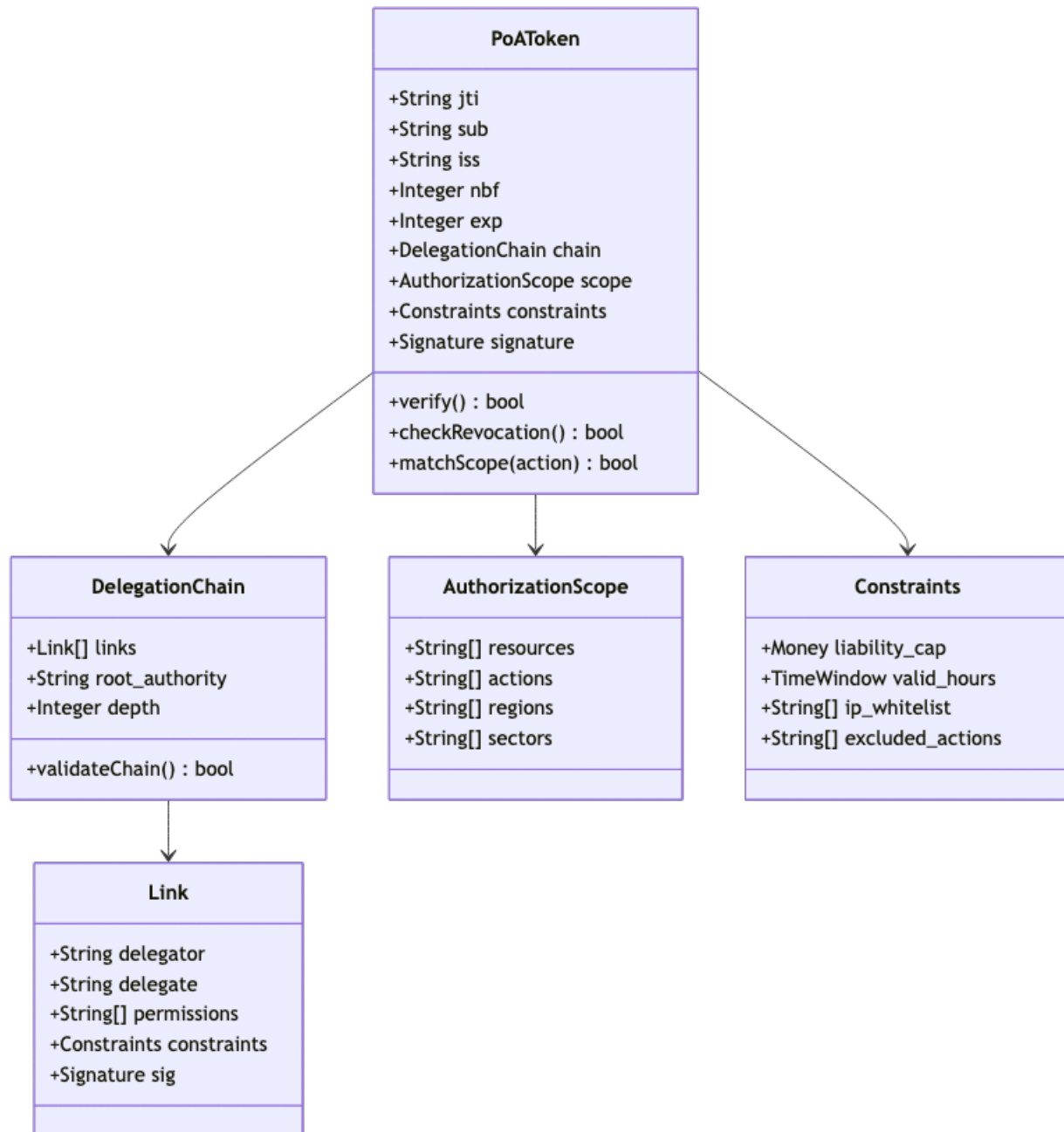


Figure 6.3: Diagram 6

6.3.2 Real-World Example: German Corporate Procurement

A German GmbH wants to authorize an AI agent for procurement. Here's the complete PoA:

```
{
  "poa_version": "2.0",
  "jti": "poa-2025-12-001-proc",

  "principal": {
    "type": "organization",
    "identity": "DE:HRB:123456",
    "name": "Mustermann GmbH",
    "jurisdiction": "DE",
    "commercial_register": {
      "authority": "Amtsgericht München",
      "registration_number": "HRB 123456",
      "registration_date": "2015-03-15"
    }
  },

  "agent": {
    "type": "autonomous_agent",
    "identity": "urn:agentauth:agent:mustermann-proc-7",
    "capability_level": "L3",
    "version": "1.2.0"
  },

  "authorization": {
    "actions": [
      "procurement:create",
      "procurement:approve",
      "payment:initiate"
    ],
    "resources": [
      "category:office_supplies",
      "category:it_equipment"
    ],
    "excluded_actions": [
      "payment:international",
      "vendor:new_registration"
    ],
    "regions": ["DE", "AT", "CH"]
  },

  "constraints": {
    "liability_cap": {
      "currency": "EUR",
      "amount": 50000,
    }
  }
}
```

```

    "period": "per_transaction"
  },
  "daily_limit": {
    "currency": "EUR",
    "amount": 200000
  },
  "approval_required_above": {
    "currency": "EUR",
    "amount": 25000
  },
  "valid_hours": {
    "timezone": "Europe/Berlin",
    "days": ["Mon", "Tue", "Wed", "Thu", "Fri"],
    "hours": "08:00-18:00"
  }
},

"chain": [
  {
    "delegator": "DE:HRB:123456:BOARD",
    "delegate": "DE:HRB:123456:CF0:Max.Mustermann",
    "authority_type": "statutory",
    "legal_basis": "GmbHG §35",
    "granted": "2024-01-15",
    "signature": "..."
  },
  {
    "delegator": "DE:HRB:123456:CF0:Max.Mustermann",
    "delegate": "urn:agentauth:agent:mustermann-proc-7",
    "authority_type": "delegated",
    "legal_basis": "Internal Delegation Policy §4.2",
    "granted": "2025-01-01",
    "constraints_narrowed": true,
    "signature": "..."
  }
],

"validity": {
  "not_before": "2025-01-01T00:00:00Z",
  "not_after": "2025-12-31T23:59:59Z"
},

"revocation_endpoint": "https://auth.mustermann.de/revocation",

"signature": {
  "algorithm": "EdDSA",
  "kid": "agent-signing-key-2025",
  "value": "..."
}

```

}
}

Example 2.2: Complete PoA for a German corporate procurement agent

This PoA enables any receiver to verify: 1. The agent is authorized by Mustermann GmbH 2. The authorization chain traces to statutory authority (GmbHG §35) 3. The transaction is within scope (office supplies, IT equipment) 4. The amount is within liability cap (€50,000 per transaction) 5. The action occurs during business hours

6.4 2.4 Pillar 3: Resilience (Degraded Mode)

What happens when the verification infrastructure fails? AgentAuth implements **graceful degradation**:

Figure 7: Architecture diagram

Figure 2.4: Resilience modes — graceful degradation under failure conditions

Mode	Conditions	Capabilities	Audit
Full	All services available	Complete verification, full scope	Real-time
Degraded	Revocation service down	Cached policies, reduced limits	Buffered
Emergency	Critical infrastructure failure	Pre-approved emergency actions only	Local
Fail Closed	No valid policy available	No authorization	Alert

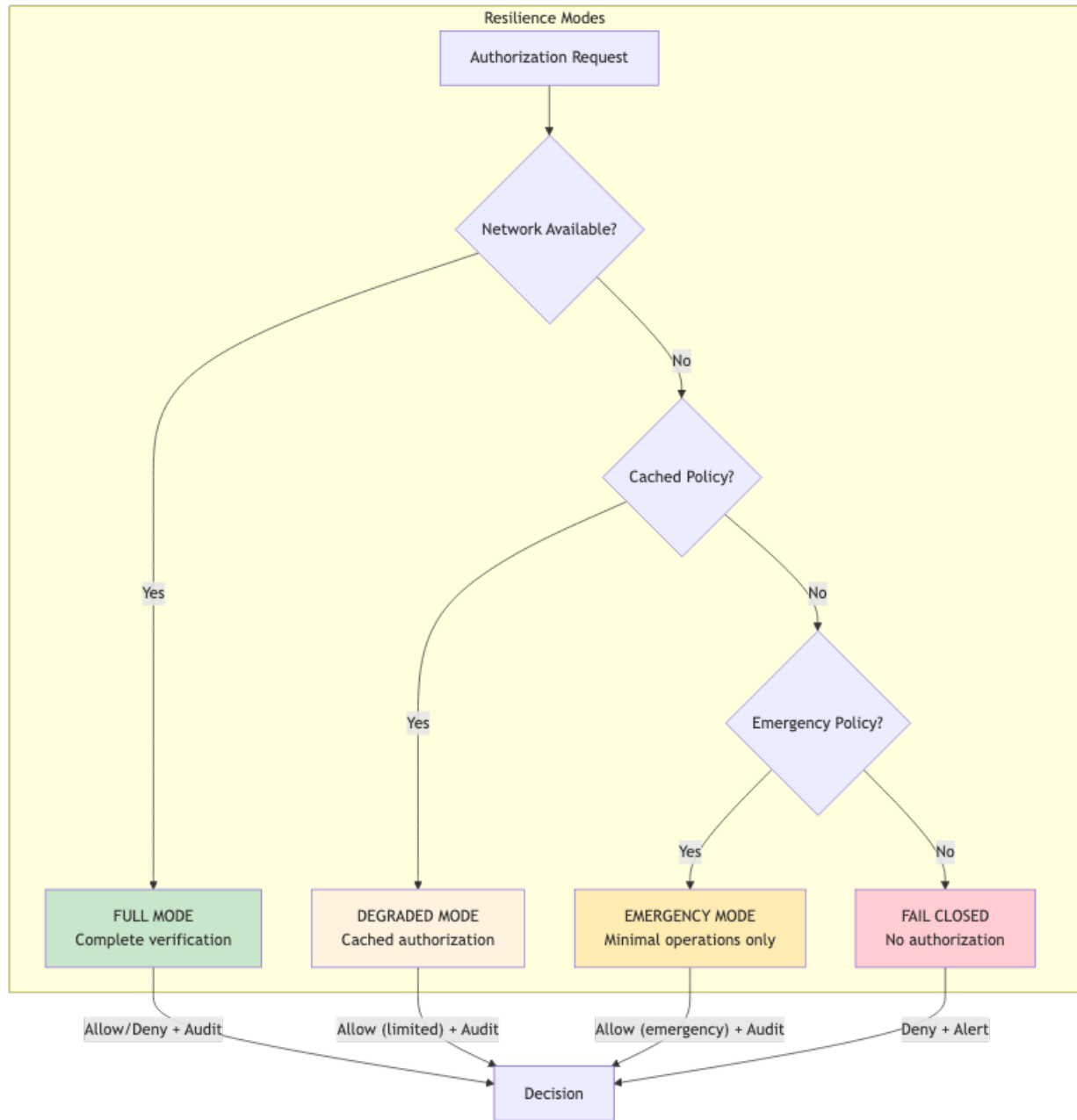


Figure 6.4: Diagram 7

Chapter 7

Chapter 3: The Technical Implementation

7.1 3.1 System Architecture

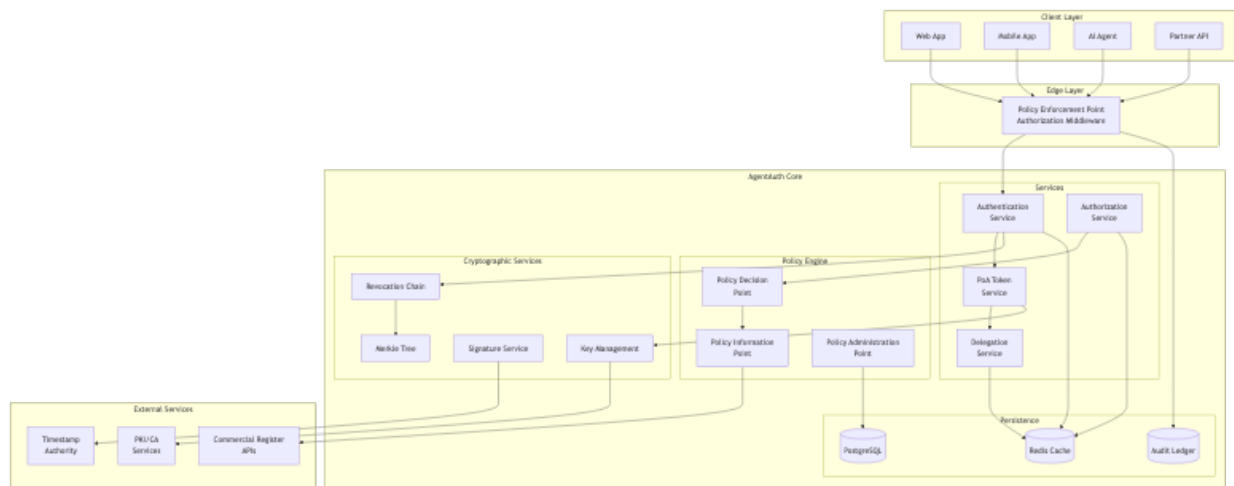


Figure 7.1: Diagram 8

Figure 8: Architecture diagram

Figure 3.1: Complete AgentAuth system architecture

7.2 3.2 Core Data Flows

7.2.1 Flow 1: PoA Token Issuance

Figure 9: Architecture diagram

Figure 3.2: PoA token issuance flow — from principal request to signed token

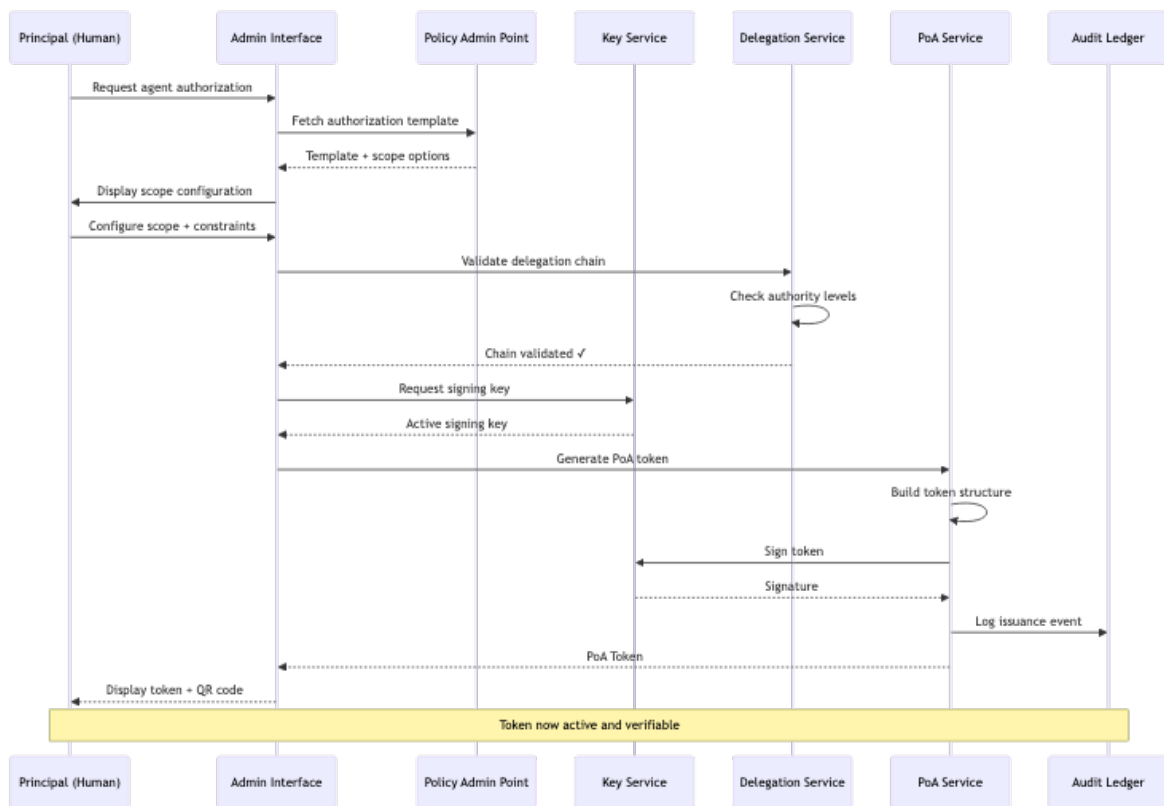


Figure 7.2: Diagram 9

7.2.2 Flow 2: Authorization Decision

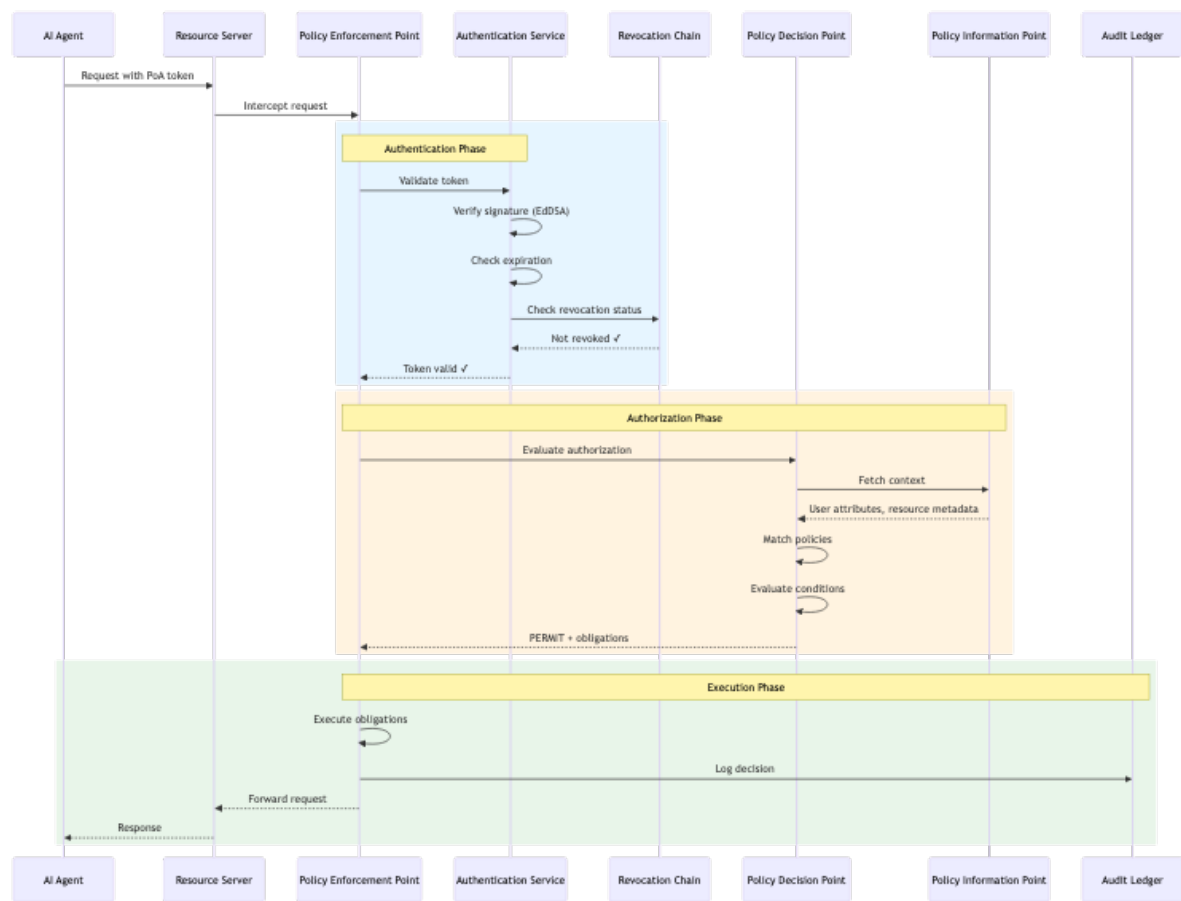


Figure 7.3: Diagram 10

Figure 10: Architecture diagram

Figure 3.3: Authorization decision flow — complete verification pipeline

7.3 3.3 Cryptographic Architecture

7.3.1 Supported Algorithms

AgentAuth implements algorithm agility to support evolving cryptographic standards:

Algorithm	Purpose	Strength	Post-Quantum
Ed25519	Signatures (default)	128-bit	
ECDSA P-256	Signatures (legacy)	128-bit	
ECDSA P-384	Signatures (high security)	192-bit	
BLS12-381	Aggregate signatures	128-bit	
Dilithium-3	Signatures (future)	192-bit	

Algorithm	Purpose	Strength	Post-Quantum
SHA-256	Hashing	128-bit	
SHA-384	Hashing (high security)	192-bit	

7.3.2 Multi-Signature Support

For high-value transactions, AgentAuth supports threshold signatures:

Figure 11: Architecture diagram

Figure 3.4: Multi-signature threshold scheme for high-value authorizations

7.4 3.4 Revocation Transparency

AgentAuth implements append-only Merkle tree-based revocation:

Figure 12: Architecture diagram

Figure 3.5: Merkle tree-based revocation with inclusion proofs

Benefits: - **Append-only:** Revocations cannot be hidden - **Verifiable:** Any party can verify status
- **Efficient:** $O(\log n)$ inclusion proofs - **Timestamped:** External anchoring proves timing

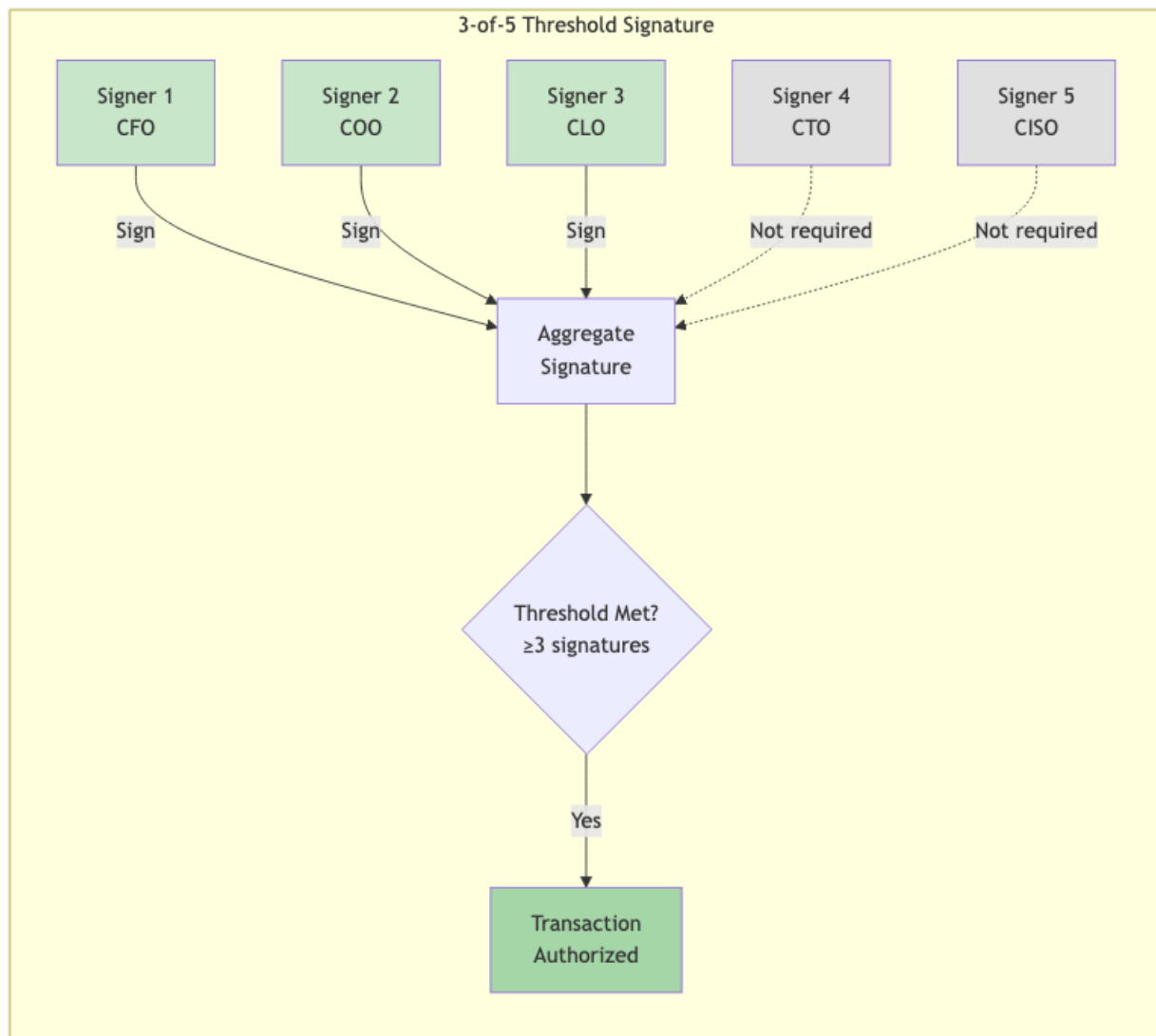


Figure 7.4: Diagram 11

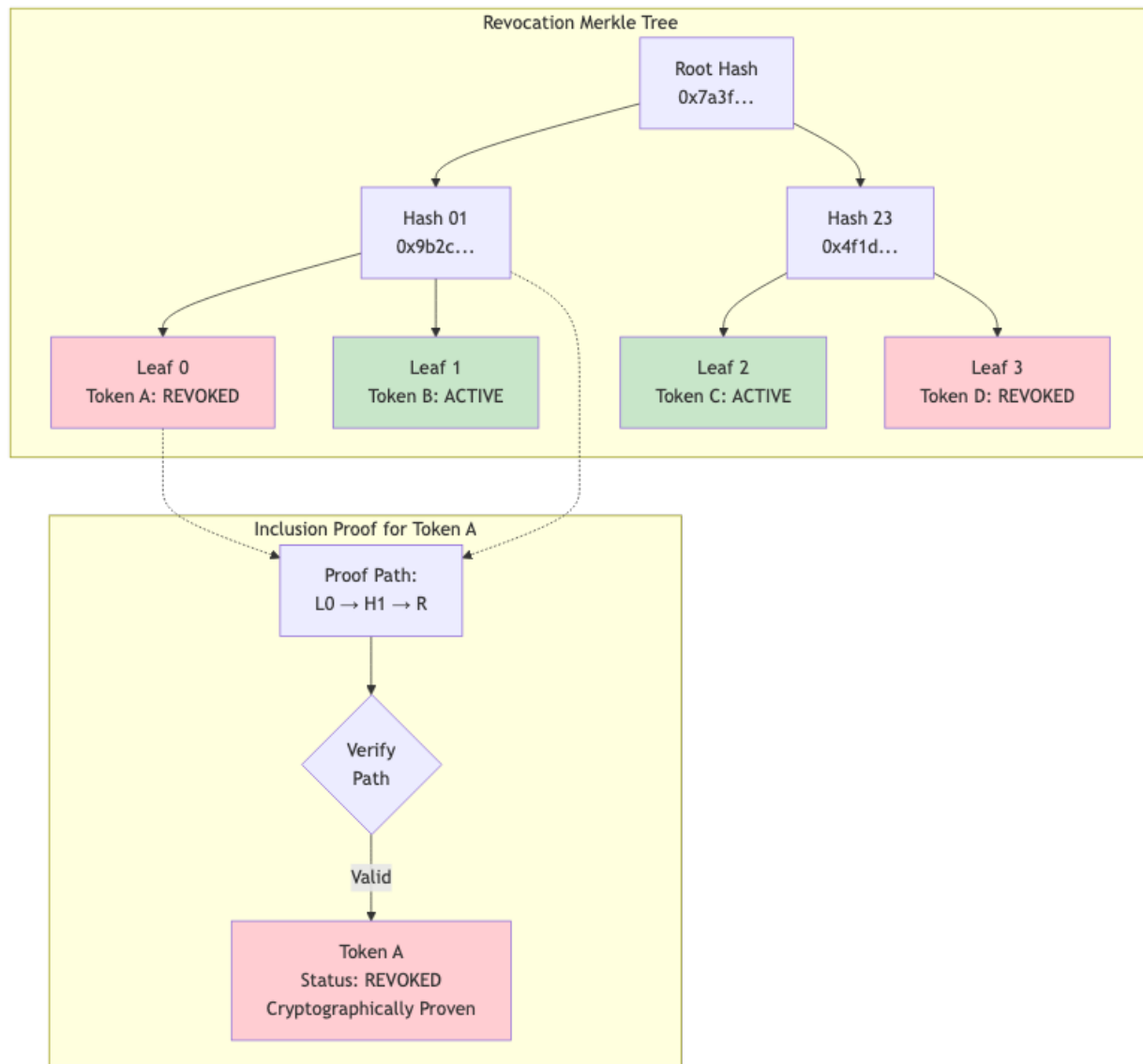


Figure 7.5: Diagram 12

Chapter 8

Chapter 4: Legal Frameworks Across Jurisdictions

8.1 4.1 The Universal Challenge

AI agents must operate across jurisdictions with different legal traditions. AgentAuth maps these differences:

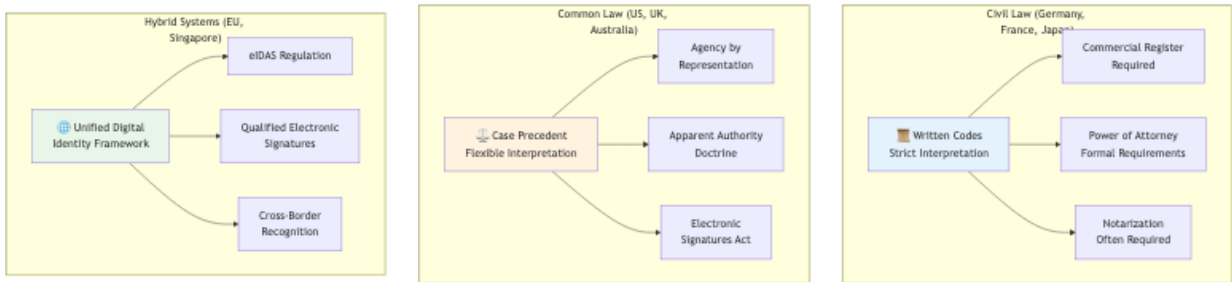


Figure 8.1: Diagram 13

Figure 13: Architecture diagram

Figure 4.1: Legal framework categories and their implications for AI authorization

8.2 4.2 Germany: The Commercial Register Model

Germany’s Commercial Register (Handelsregister) provides a template for verifiable corporate authority.

8.2.1 German Implementation Pattern

Figure 14: Architecture diagram

Figure 4.2: German commercial register integration flow

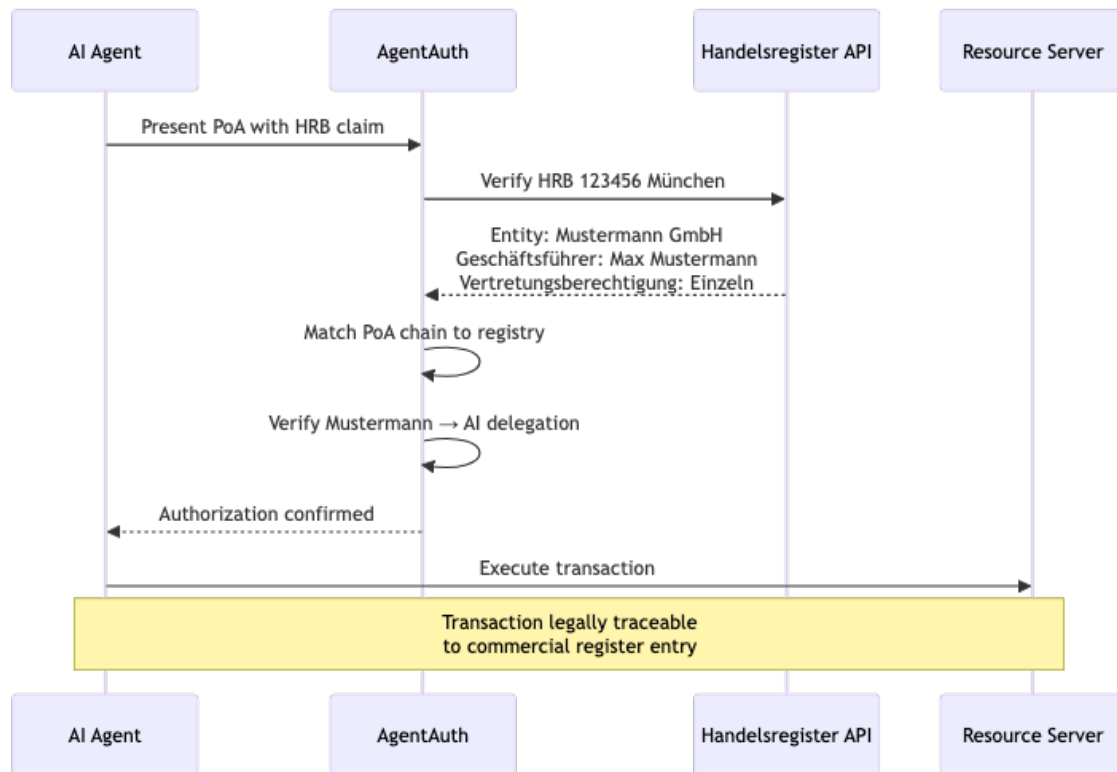


Figure 8.2: Diagram 14

8.2.2 Key Legal Provisions

Provision	Implication for AI Agents
§164 BGB	Agent acts in name of principal
§35 GmbHG	Managing directors represent company
§49 HGB	<i>Prokura</i> (commercial power of attorney)
§166 BGB	Knowledge imputed to principal

8.3 4.3 United States: The Apparent Authority Model

US law recognizes “apparent authority” — if a third party reasonably believes an agent is authorized, the principal may be bound.

8.3.1 Implications for AgentAuth

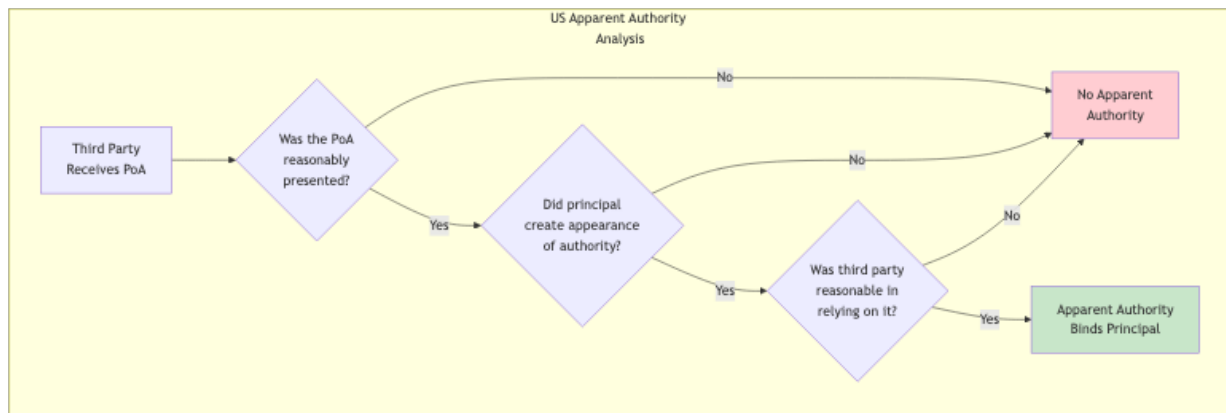


Figure 8.3: Diagram 15

Figure 15: Architecture diagram

Figure 4.3: US apparent authority doctrine analysis

This creates a **higher bar for AgentAuth verification** in US contexts: - Principals must explicitly define agent scope - Scope must be verifiable by third parties - Over-broad grants create liability risk

8.4 4.4 European Union: The eIDAS Framework

The EU’s eIDAS Regulation provides a unified framework for electronic identification and trust services.

8.4.1 eIDAS Trust Levels

Level	Verification	AgentAuth Equivalent
Low	Self-asserted	Entity Profile (unsigned)
Substantial	Identity verified	Entity Profile + KYC
High	In-person + qualified cert	Entity Profile + QES

8.4.2 AgentAuth eIDAS Integration

```
{
  "identity_verification": {
    "method": "eIDAS",
    "trust_level": "substantial",
    "issuing_country": "DE",
    "verification_service": "urn:idas:de:bundesnetzagentur",
    "verification_date": "2025-01-15T10:30:00Z",
    "verification_proof": {
      "assertion_id": "idas-de-2025-001234",
      "signature": "..."
    }
  }
}
```

Example 4.1: eIDAS identity verification integrated into AgentAuth entity profile

Chapter 9

Chapter 5: Building Trustworthy Agents

9.1 5.1 The Agent Capability Levels

AgentAuth defines capability levels based on autonomy:

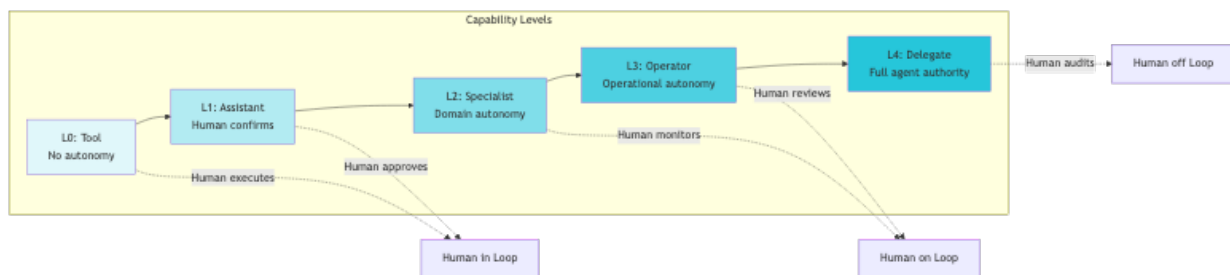


Figure 9.1: Diagram 16

Figure 16: Architecture diagram

Figure 5.1: Agent capability levels and human oversight requirements

9.1.1 Capability Level Requirements

Level	Authorization	Constraints	Example
L0	None required	N/A	Calculator tool
L1	Scope-limited	All actions approved	Email draft assistant
L2	Domain PoA	Domain boundaries	Customer support bot
L3	Full PoA + limits	Financial/temporal limits	Procurement agent
L4	Full PoA + audit	Regulatory supervision	Trading algorithm

9.2 5.2 Implementing a Procurement Agent

9.2.1 Step 1: Define the Entity Profile

```
// Create entity profile for procurement agent
profile := &agentauth.EntityProfile{
    EntityID: "urn:agentauth:agent:acme-proc-001",
    Type:     agentauth.EntityTypeAgent,
    PublicKey: agentauth.PublicKey{
        Algorithm: "Ed25519",
        KeyBytes:   publicKeyBytes,
    },
    LegalMetadata: agentauth.LegalMetadata{
        Organization: agentauth.Organization{
            Name:         "Acme Corporation",
            Jurisdiction: "DE",
            RegisterID:   "HRB 123456",
        },
        CapabilityLevel: agentauth.CapabilityL3,
        LiabilityCap:     agentauth.Money{Amount: 1000000, Currency: "EUR"},
    },
    Validity: agentauth.Validity{
        NotBefore: time.Now(),
        NotAfter:   time.Now().AddDate(1, 0, 0),
    },
}
```

Example 5.1: Entity profile creation in Go

9.2.2 Step 2: Issue PoA Token

```
// Build PoA for procurement operations
poa := &agentauth.PoA{
    JTI:   uuid.New().String(),
    Sub:   "urn:agentauth:agent:acme-proc-001",
    Iss:   "urn:agentauth:issuer:acme-auth",
    Chain: agentauth.DelegationChain{
        Links: []agentauth.DelegationLink{
            {
                Delegator: "DE:HRB:123456:BOARD",
                Delegate:  "DE:HRB:123456:CFO",
                Authority: agentauth.AuthorityStatutory,
                Signature: boardSignature,
            },
            {
                Delegator: "DE:HRB:123456:CFO",
                Delegate: "urn:agentauth:agent:acme-proc-001",
                Authority: agentauth.AuthorityDelegated,
                Scope: agentauth.Scope{

```

```

        Resources: []string{"procurement:*"},
        Actions:   []string{"create", "approve", "payment:domestic"},
    },
    Constraints: agentauth.Constraints{
        LiabilityCap: agentauth.Money{Amount: 50000, Currency: "EUR"},
        ValidHours:   []string{"Mon-Fri 08:00-18:00 Europe/Berlin"},
    },
    Signature: cfoSignature,
},
},
Validity: agentauth.Validity{
    NotBefore: time.Now(),
    NotAfter:  time.Now().AddDate(0, 3, 0), // 3 months
},
}

```

// Sign the PoA

```
signedPoA, err := poaService.Sign(poa, signingKey)
```

Example 5.2: PoA token issuance with delegation chain

9.2.3 Step 3: Execute Authorized Transaction

```

// Agent executes procurement
func (agent *ProcurementAgent) PlaceOrder(order *Order) error {
    // Attach PoA to request
    ctx := agentauth.WithPoA(context.Background(), agent.poaToken)

    // Call procurement API
    resp, err := agent.procurementClient.CreateOrder(ctx, &CreateOrderRequest{
        Vendor:      order.Vendor,
        Items:       order.Items,
        TotalAmount: order.TotalAmount,
    })

    if err != nil {
        // Handle authorization failure
        if errors.Is(err, agentauth.ErrAuthorizationDenied) {
            return fmt.Errorf("order exceeds agent authority: %w", err)
        }
        return err
    }

    return nil
}

```

Example 5.3: Agent executing an authorized transaction

9.3 5.3 Verification Workflow

The receiving system verifies the PoA:

```
// Verification middleware
func VerifyPoA(next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        // Extract PoA from request
        poa, err := agentauth.ExtractPoA(r)
        if err != nil {
            http.Error(w, "Missing PoA", http.StatusUnauthorized)
            return
        }

        // Comprehensive verification
        result, err := verifier.Verify(r.Context(), poa, &agentauth.VerifyOptions{
            CheckRevocation:    true,
            CheckChainValidity:  true,
            CheckConstraints:    true,
            RequiredScope: &agentauth.Scope{
                Resources: []string{"procurement:orders"},
                Actions:    []string{"create"},
            },
            TransactionContext: &agentauth.TransactionContext{
                Amount:    extractAmount(r),
                Currency: "EUR",
            },
        })

        if err != nil || !result.Valid {
            auditLog.Warn("PoA verification failed",
                "poa_id", poa.JTI,
                "reason", result.DenialReason,
            )
            http.Error(w, "Authorization denied", http.StatusForbidden)
            return
        }

        // Attach verification result to context
        ctx := agentauth.WithVerification(r.Context(), result)
        next.ServeHTTP(w, r.WithContext(ctx))
    })
}
```

Example 5.4: PoA verification middleware

Chapter 10

Chapter 6: Operational Excellence

10.1 6.1 Observability

AgentAuth provides comprehensive observability:



Figure 10.1: Diagram 17

Figure 17: Architecture diagram

Figure 6.1: AgentAuth observability architecture

10.1.1 Key Metrics

Metric	Type	Description
agentauth_poa_issued_total	Counter	PoA tokens issued
agentauth_authorization_decisions_total	Counter	Authorization decisions (by outcome)
agentauth_verification_latency_seconds	Histogram	Verification latency distribution
agentauth_revocation_chain_depth	Gauge	Current revocation chain depth
agentauth_delegation_chain_depth	Gauge	Delegation chain depth distribution
agentauth_constraint_violations_total	Counter	Constraint violations (by type)

10.2 6.2 Audit & Compliance

10.2.1 Cryptographic Audit Trail

Every authorization decision is recorded in an append-only audit ledger:

```

{
  "audit_id": "aud-2025-12-30-001234",
  "timestamp": "2025-12-30T14:23:45.123Z",
  "event_type": "AUTHORIZATION_DECISION",
  "decision": "PERMIT",
  "request": {
    "poa_id": "poa-2025-001",
    "action": "procurement:create",
    "resource": "orders/PO-2025-9876",
    "amount": {"value": 45000, "currency": "EUR"}
  },
  "verification": {
    "chain_valid": true,
    "chain_depth": 2,
    "revocation_checked": true,
    "constraints_satisfied": true
  },
  "context": {
    "source_ip": "10.0.1.50",
    "user_agent": "AcmeProcBot/1.2.0"
  },
  "digest": "sha256:7a3f9b2c4d5e6f...",
  "previous_digest": "sha256:4f1d8e3a2b6c...",
  "signature": "..."
}

```

Example 6.1: Cryptographic audit entry

10.2.2 External Anchoring

For regulatory compliance, audit entries are anchored to external timestamping authorities:

Figure 18: Architecture diagram

Figure 6.2: External audit anchoring flow

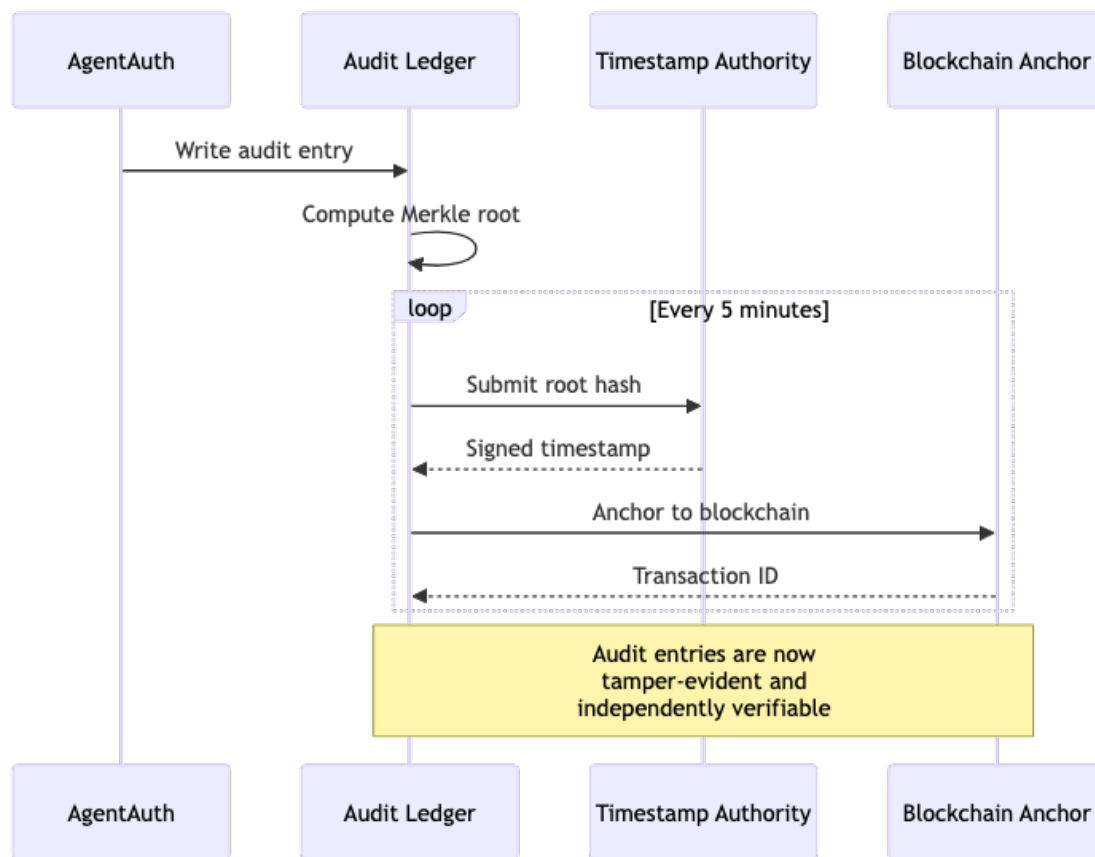


Figure 10.2: Diagram 18

Chapter 11

Chapter 7: Advanced Patterns

11.1 7.1 Multi-Party Authorization

Some transactions require multiple approvals:

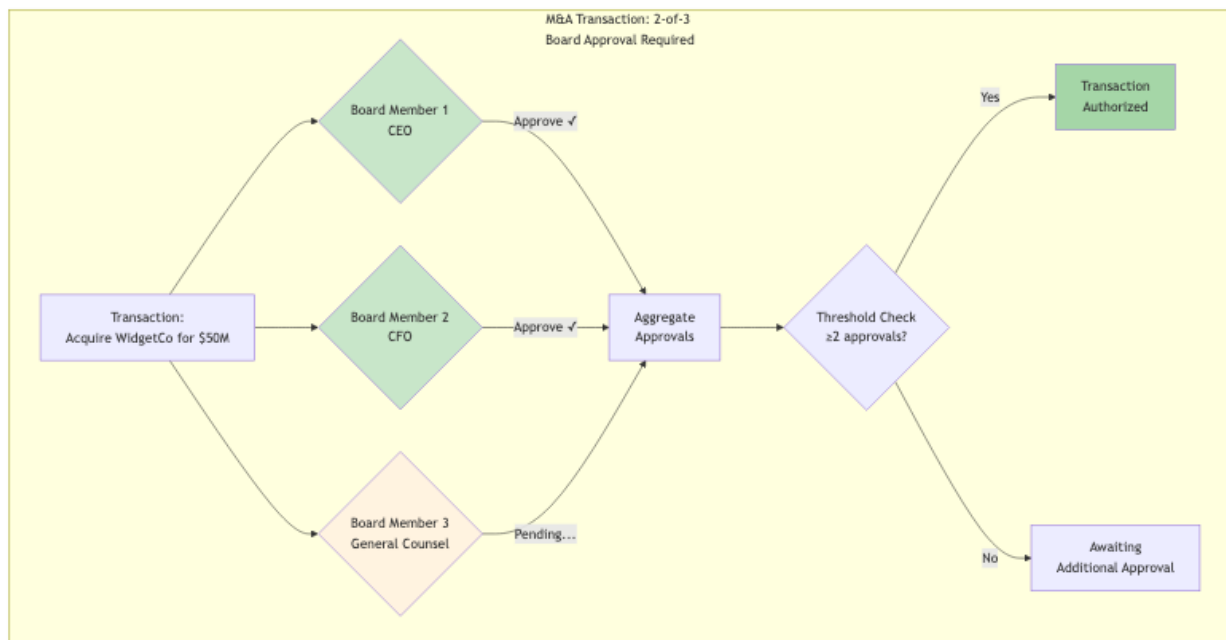


Figure 11.1: Diagram 19

Figure 19: Architecture diagram

Figure 7.1: Multi-party authorization with threshold approval

11.2 7.2 Conditional Delegation

Delegations can include dynamic conditions:

```
{
  "constraints": {
    "conditions": [
      {
        "type": "market_price",
        "asset": "AAPL",
        "operator": "less_than",
        "value": 200.00,
        "oracle": "urn:agentauth:oracle:nasdaq"
      },
      {
        "type": "risk_score",
        "operator": "less_than",
        "value": 0.7,
        "oracle": "urn:agentauth:oracle:risk-engine"
      }
    ],
    "condition_logic": "ALL"
  }
}
```

Example 7.1: Conditional delegation with external oracles

11.3 7.3 Cascade Revocation

When a delegation is revoked, all derived delegations are automatically revoked:

Figure 20: Architecture diagram

Figure 7.2: Cascade revocation — revoking Bob’s authority revokes all derived delegations

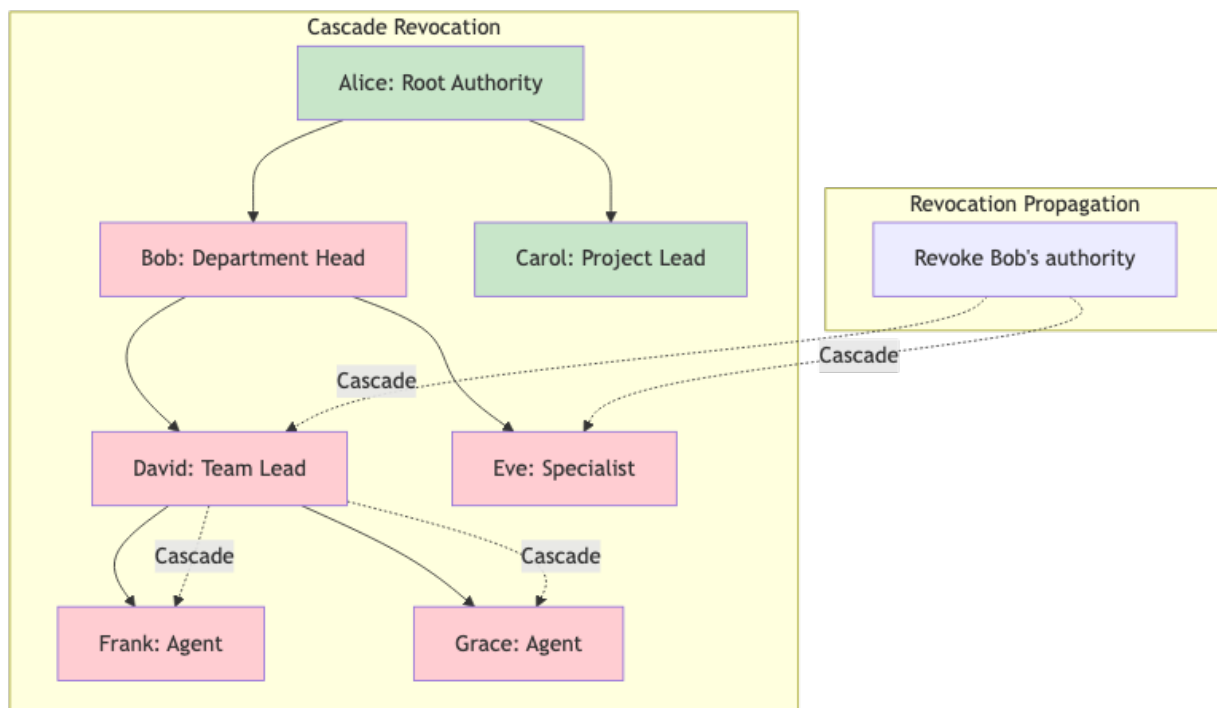


Figure 11.2: Diagram 20

Chapter 12

Appendix A: Quick Reference

12.1 A.1 PoA Token Claims

Claim	Type	Required	Description
jti	String		Unique token identifier
sub	String		Agent identity
iss	String		Token issuer
nbf	Integer		Not before (Unix timestamp)
exp	Integer		Expiration (Unix timestamp)
chain	Array		Delegation chain
scope	Object		Authorization scope
constraints	Object		Optional constraints

12.2 A.2 Error Codes

Code	HTTP Status	Description
invalid_token	401	Token signature invalid
token_expired	401	Token has expired
token_revoked	401	Token has been revoked
insufficient_scope	403	Action not in scope
constraint_violated	403	Constraint not satisfied
chain_invalid	403	Delegation chain broken

12.3 A.3 API Endpoints

Endpoint	Method	Description
/api/v1/poa/issue	POST	Issue new PoA
/api/v1/poa/verify	POST	Verify PoA
/api/v1/revocation/revoke	POST	Revoke PoA

Endpoint	Method	Description
/api/v1/revocation/status	GET	Check revocation status
/api/v1/chain/validate	POST	Validate delegation chain

Chapter 13

Appendix B: Glossary

Term	Definition
PoA	Proof of Authorization — cryptographically signed token proving agent authority
Delegation Chain	Sequence of signed authorizations from root authority to agent
AAP-001	Agent Authorization Protocol — Core identity and authentication
AAP-002	Agent Authorization Protocol — Delegation and PoA tokens
PEP	Policy Enforcement Point — component that enforces authorization decisions
PDP	Policy Decision Point — component that evaluates authorization policies
PIP	Policy Information Point — component that provides context attributes

Chapter 14

About the AgentAuth Project

AgentAuth is an open-source authorization framework for autonomous AI agents.

- **Repository:** github.com/mauriciomferz/Gauth_go
- **Version:** 1.0.0 (December 2025)
- **License:** MIT
- **Conformance:** AAP-001 (100%), AAP-002 (100%)

14.1 Contributors

This framework represents hundreds of engineering hours across cryptography, distributed systems, and legal compliance domains.

“The agent’s signature is not just a cryptographic value. It is the bridge between silicon and law, between algorithm and accountability. When we sign, we accept responsibility. When agents sign, civilization scales.”

END OF MANUSCRIPT

